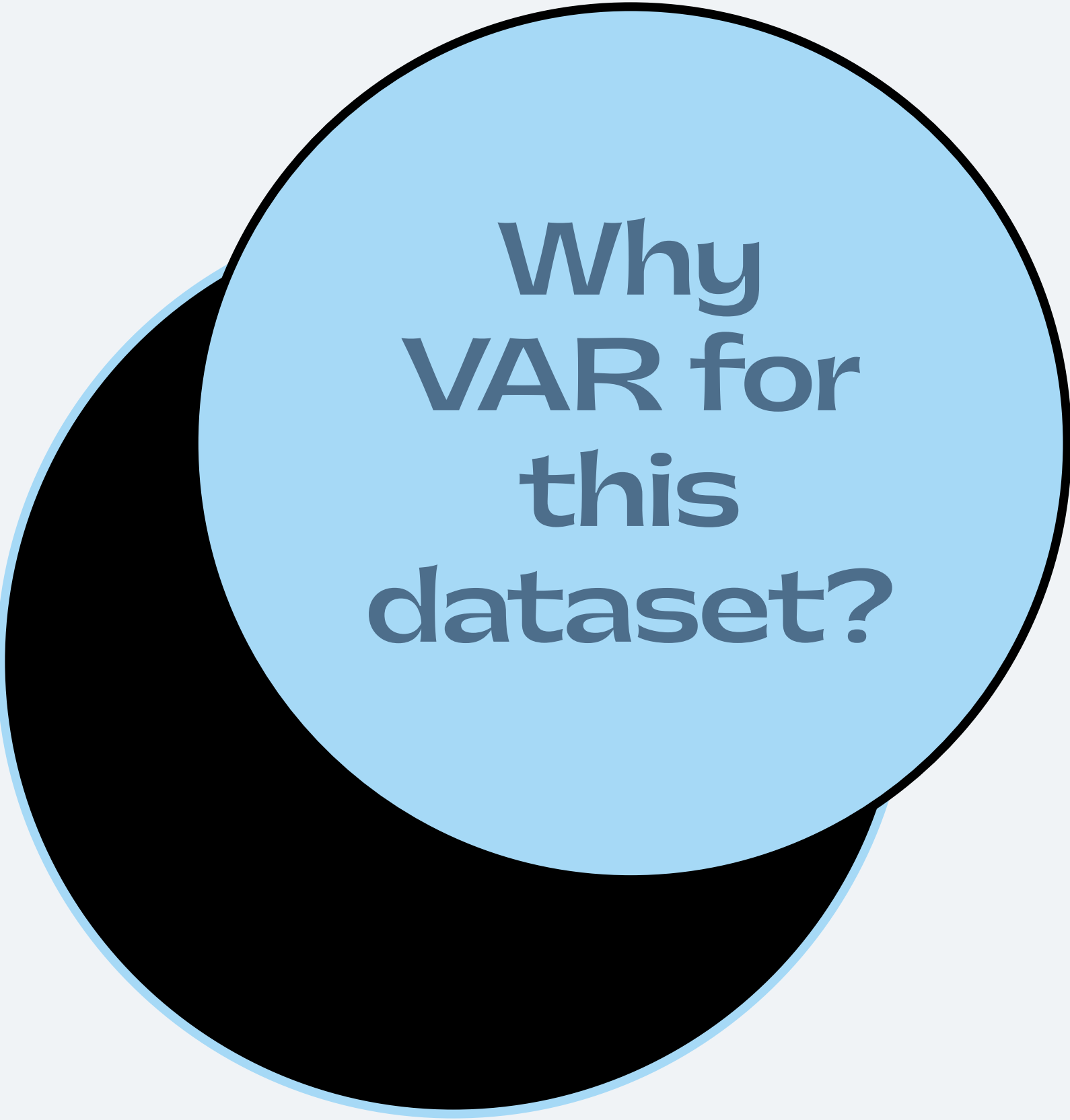


UNDERSTANDING ECONOMIC DYNAMICS

VAR Analysis of Monetary Policy

DS809 Final Project





Why VAR for this dataset?

- All three variables influence each other over time
- ARIMA: models one series at a time, ignores feedback
- Dynamic regression: treats EFR as exogenous
- VAR: treats all variables as endogenous, lets them predict each other

What is VAR?

- Multivariate extension of AR(p)
- Each variable regressed on lags of all variables
- Lag order chosen by information criteria (AIC/BIC)
- Key tools: impulse responses, Granger causality, variance decomposition

The Univariate Issue

- Unemployment alone misses Fed reaction to labor market
- Policy decisions respond to unemployment and inflation jointly
- Need a system that allows mutual influence, not one-way effects

Our Dataset



Total record

752 monthly
observations: July 1954
– February 2017



Variables

EFFR, Unemployment
Rate, Inflation Rate



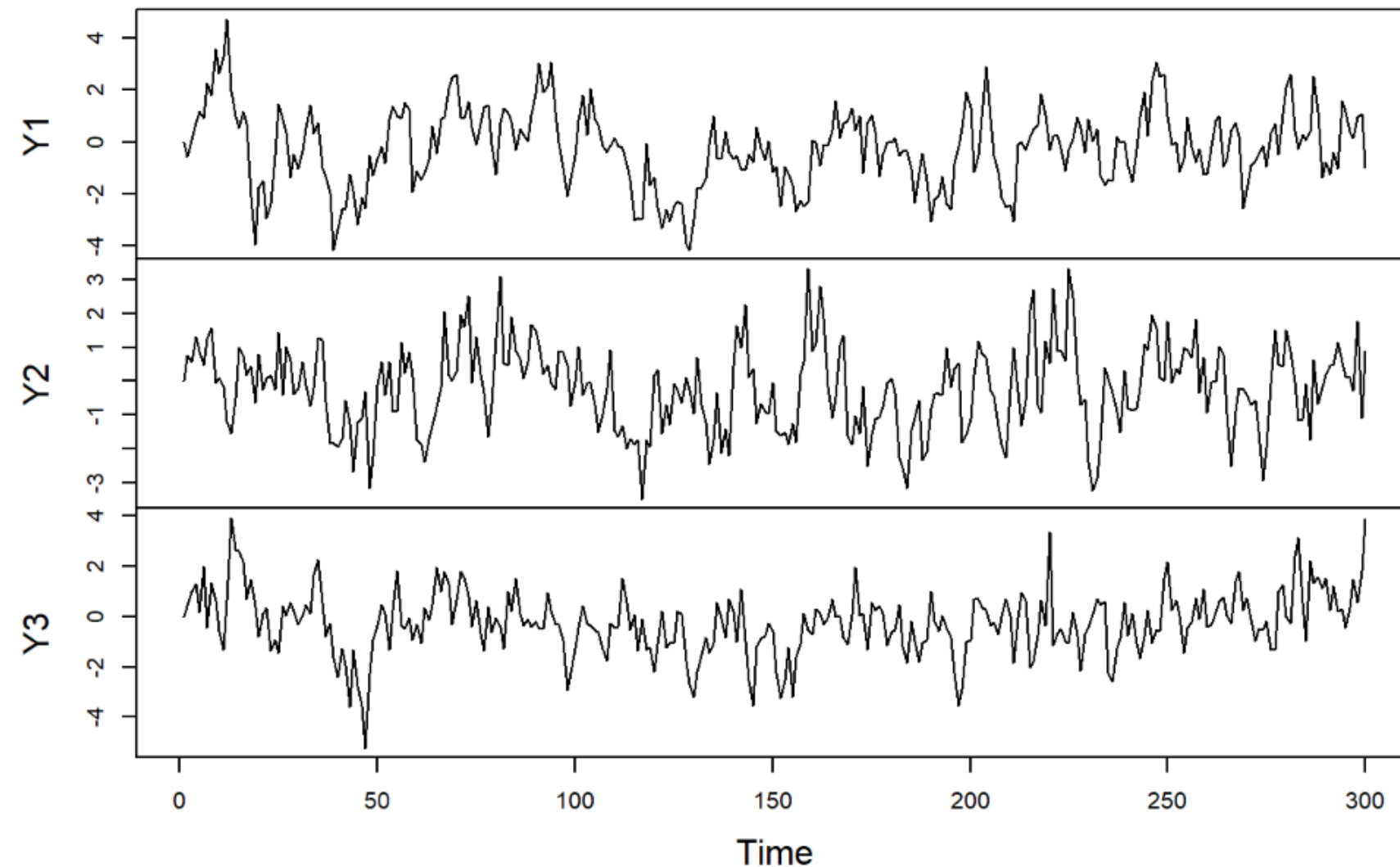
Missing values

Missing values handled via linear
interpolation

Toy Example

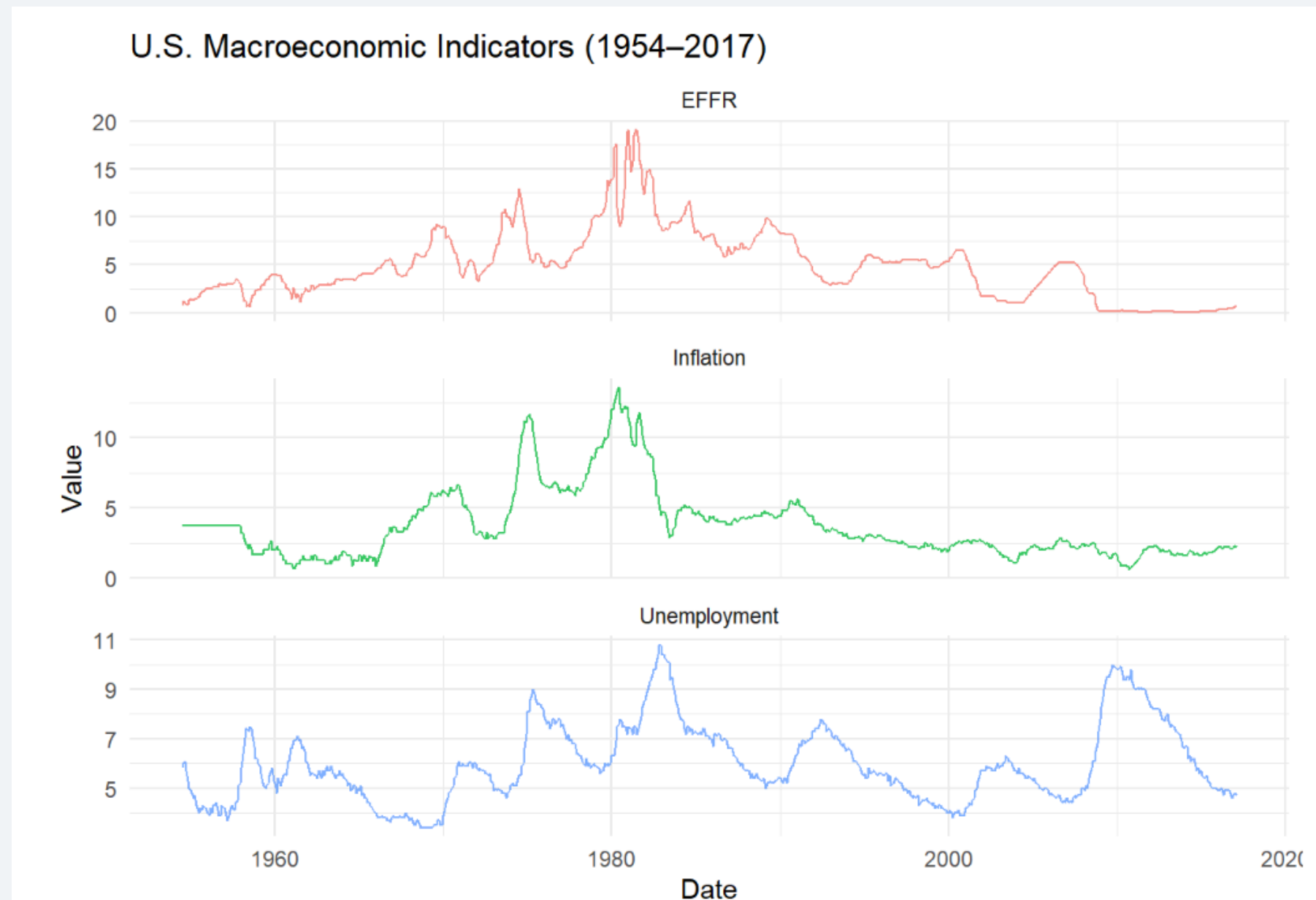
- Simulate 3-variable VAR(1) system with a known 3×3 coefficient matrix Φ .
- Fit VAR(1) to the simulated data and estimate Φ .
- **Result:** estimated coefficients are very close to the true Φ (correctly recover strong self-persistence and cross-effects).
- Takeaway: the VAR procedure can **recover the underlying structure** in a controlled setting before we apply it to U.S. macro data.

Toy Example: Simulated Trivariate VAR(1) Process



Exploratory Data Analysis

- Clear co-movement visible across the three series
- 1980s: large EFFR hikes followed by higher unemployment
- 1990s: declining rates, unemployment, and inflation together
- Visual motivation for using a multivariate model

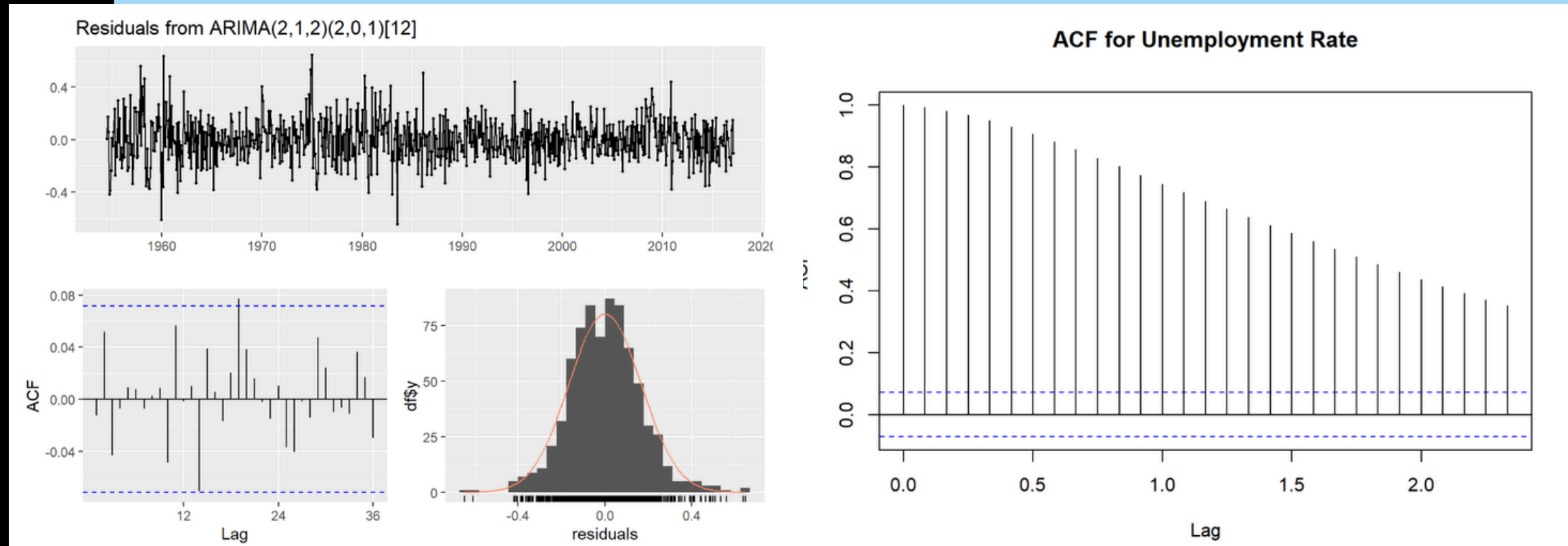


Stationarity Testing

- Stationarity checked with ADF and KPSS tests
- Evidence of unit roots in EFFR and inflation
- Series made stationary using first differences
- Differenced series satisfy stationarity assumptions
- After differencing, the series are stable enough for VAR modeling.

	Variable	ADF_stat	ADF_pval	ADF_stationary	KPSS_stat
Dickey-Fuller	Unemployment	-3.564	0.0361	Yes (reject H0)	0.950
Dickey-Fuller1	EFFR	-2.919	0.1892	No (fail to reject)	2.503
Dickey-Fuller2	Inflation	-3.237	0.0815	No (fail to reject)	2.155
	KPSS_pval	KPSS_stationary			
Dickey-Fuller	0.01	No (reject H0)			
Dickey-Fuller1	0.01	No (reject H0)			
Dickey-Fuller2	0.01	No (reject H0)			

	Variable	ADF_stat	p_value	Stationary
Dickey-Fuller	dUnemployment	-7.445	0.01	Yes
Dickey-Fuller1	dEFFR	-8.221	0.01	Yes
Dickey-Fuller2	dInflation	-6.351	0.01	Yes



VAR Model Selection & Fit

- Lag order chosen with VARselect: BIC $\rightarrow p = 3$
- Final model: VAR(3) on differenced series
- High R^2 values for all three equations
- Joint F-tests show overall model significance
- VAR(3) balances model fit and parsimony (enough lags to capture dynamics without overfitting)

Granger Causality Tests

Granger causality H0: EFR do not Granger-cause Unemployment Inflation

data: VAR object fit_var

F-Test = 8.8874, df1 = 6, df2 = 2214, p-value = 1.324e-09

Granger causality H0: Unemployment do not Granger-cause EFR Inflation

data: VAR object fit_var

F-Test = 7.2636, df1 = 6, df2 = 2214, p-value = 1.066e-07

Granger causality H0: Inflation do not Granger-cause Unemployment EFR

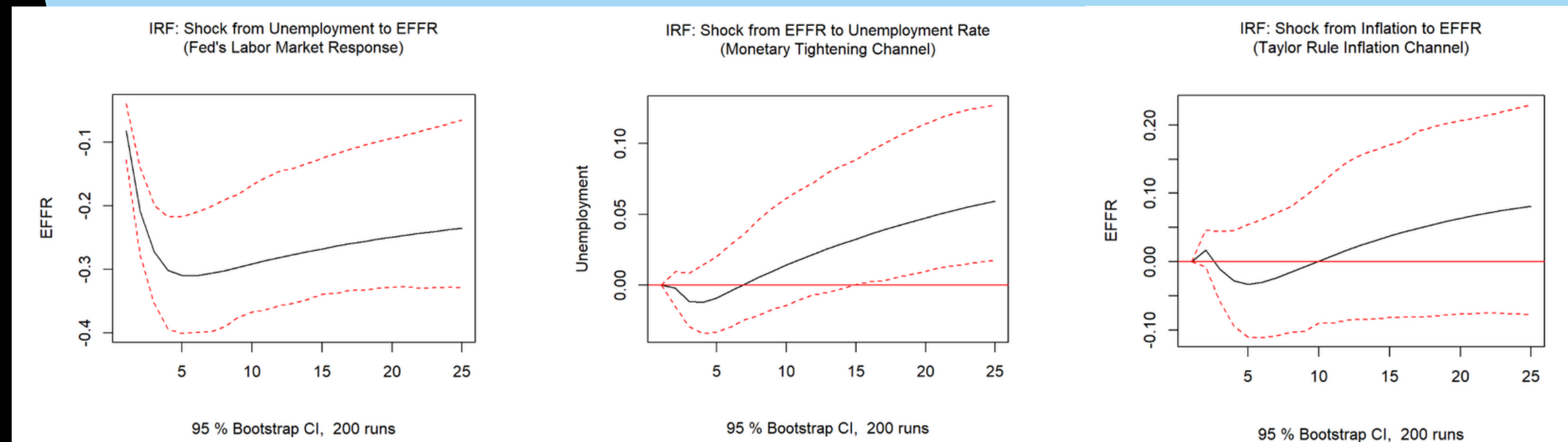
data: VAR object fit_var

F-Test = 4.0674, df1 = 6, df2 = 2214, p-value = 0.0004614

- EFR Granger-causes unemployment (rate hikes precede job losses)
- Unemployment Granger-causes EFR (weak labor markets precede cuts)
- Inflation Granger-causes EFR (higher inflation precedes rate hikes)
- Evidence of strong bidirectional feedback in the system

Result

Monetary policy and the real economy jointly influence each other, not just one direction

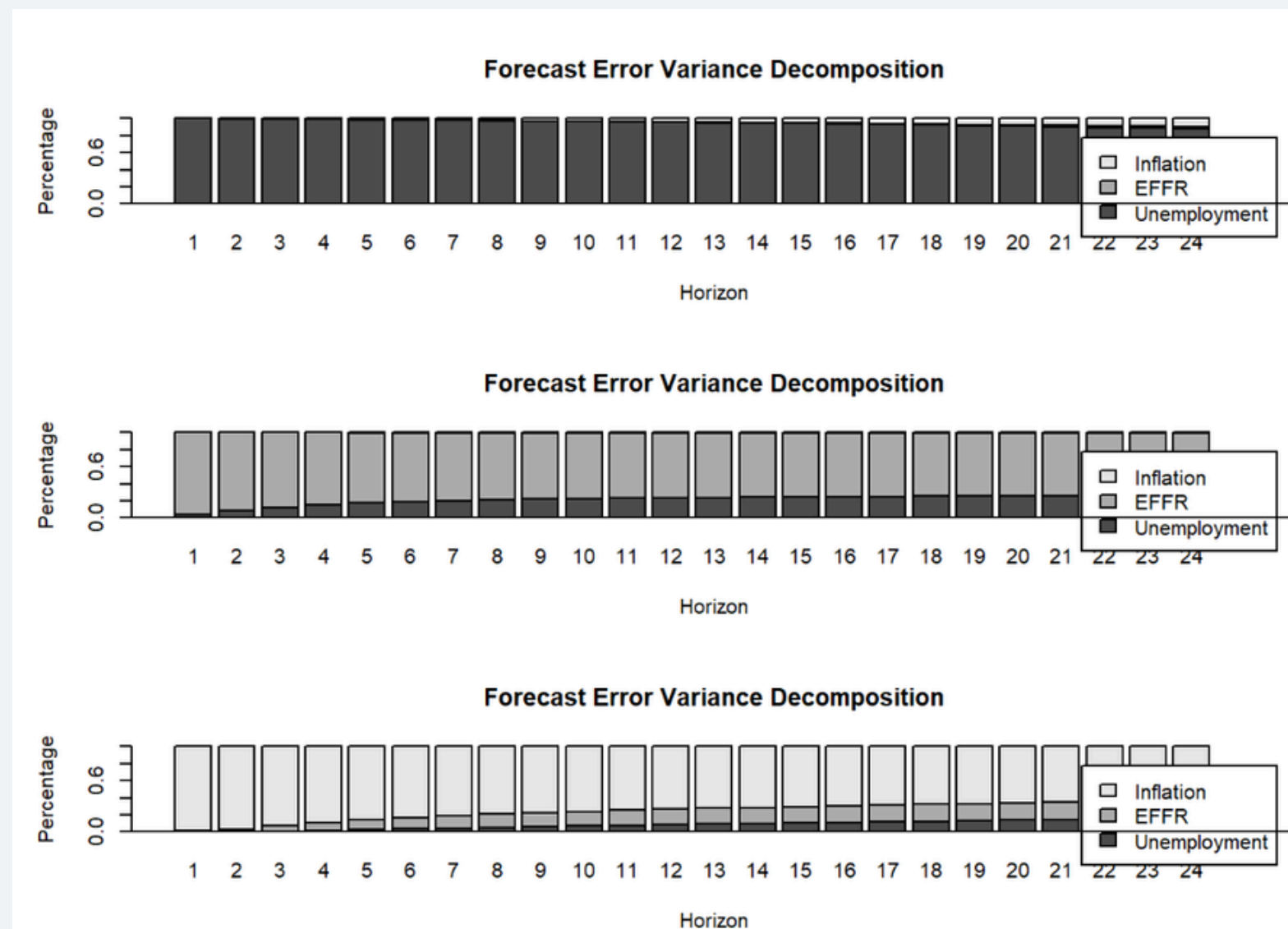


Impulse Response Functions (IRF)

- EFR shock to unemployment rises gradually over several months
- Unemployment shock to Fed lowers EFR with a lag
- Inflation shock to Fed raises EFR, then response fades over time
- Effects are delayed and persistent, not immediate

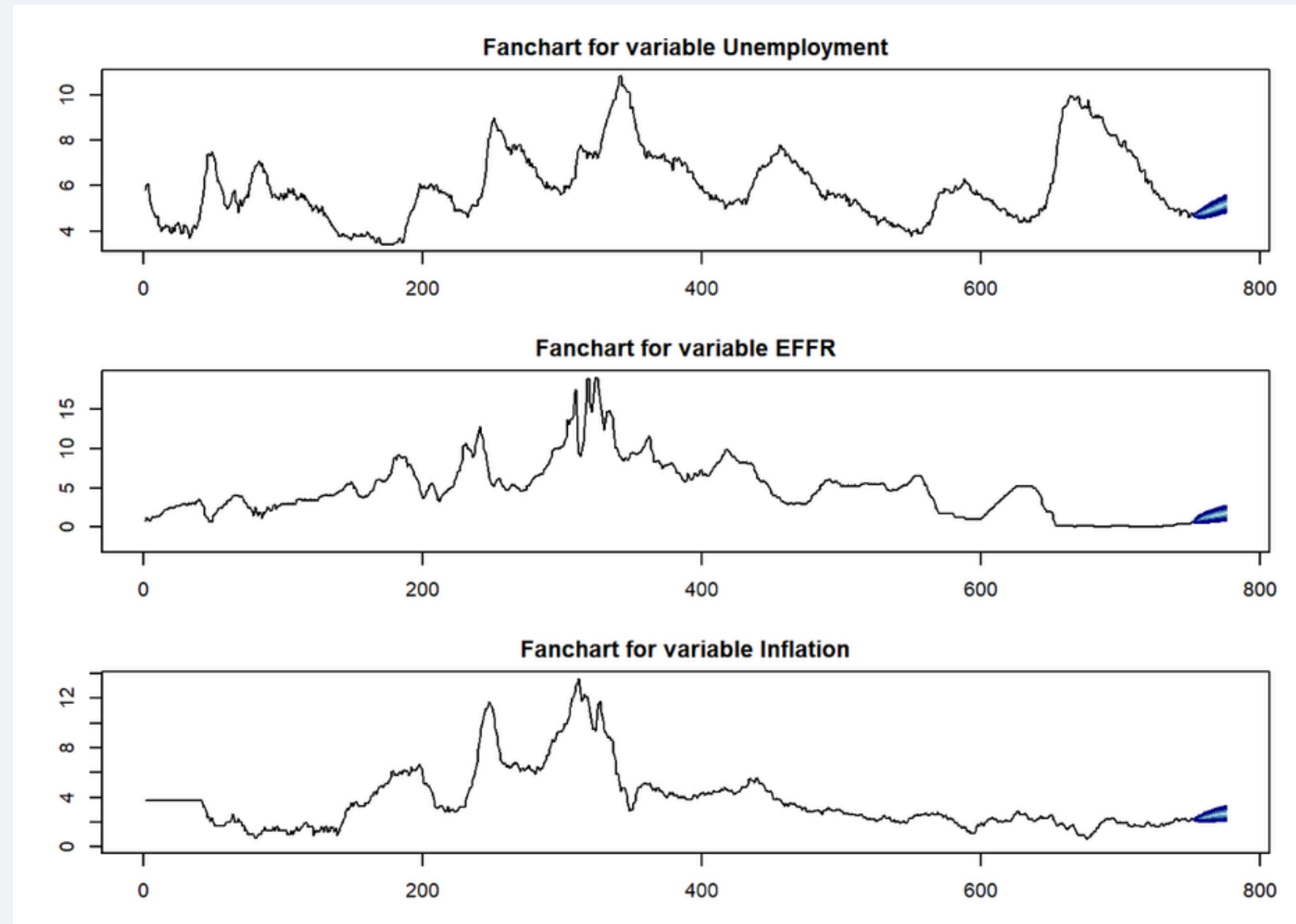
Forecast Error Variance Decomposition

- At 1 month, each variable is almost entirely self-driven
- By 24 months, other variables explain a growing share
- Unemployment: inflation explains ~10% of its variance at 24 months
- EFR: unemployment & inflation together explain ~30%+ of its variance
- Inflation: most externally influenced
EFR & unemployment explain ~40%+
- Policy effects are slow-building, not instant



Multivariate forecasting

- Joint forecasts for unemployment, EFR, and inflation
- Forecast paths reflect historical co-movement
- Useful for scenario analysis under policy shocks
- Forecasts move together in economically sensible ways, respecting historical co-movement.



Conclusion

- VAR confirms strong two-way links among EFFR, unemployment, and inflation.
- Granger tests, IRFs, and FEVD support a meaningful policy–real economy channel.
- Unemployment is highly persistent; policy effects build over longer horizons.

Pros

- Uses a multivariate framework that respects feedback between policy and the economy.
- Long monthly sample (1954–2017) gives good power to detect dynamic relationships.
- Rich toolkit (Granger, IRF, FEVD) provides interpretable policy insights.

Cons

- Assumes constant coefficients, so structural breaks and regime shifts may be missed.
- Does not explicitly model cointegration or long-run equilibrium relationships.
- Results depend on differencing and lag choice; alternative specifications might change details.